Word Skipping in L1 and L2 English Readers: A Multifaceted Analysis of Parafoveal Processing

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Abstract

This study investigates how oculomotor and linguistic factors shape first-pass word skipping in L1 and L2 English readers, with a focus on parafoveal processing. Eye-tracking data from 190 participants (L1: 88, L2: 102) were analyzed using a logistic linear mixed-effects model to assess the effects of word length, launch site distance, word frequency, and word predictability (GPT-2 surprisal), as well as interactions among these predictors and group differences. Across readers, shorter words, higher-frequency words, and more predictable words showed higher skipping probabilities. Crucially, parafoveal preview, indexed by launch site distance, substantially modulated all other factors: length, frequency and predictability effects were strongest when parafoveal preview quality was high (short launch site distance). L1 and L2 readers differed systematically in their reliance on these factors. L1 readers demonstrated greater skipping rate overall and stronger sensitivity to word frequency, reflecting efficient integration of linguistic knowledge. In contrast, L2 readers were particularly sensitive to low-level oculomotor cues, showed weaker frequency effects, and more strongly influenced by parafoveal preview quality when words were short or highly frequent. These findings support a hybrid account in which oculomotor and linguistic factors jointly determine skipping behavior, while L1 and L2 readers differ in how they weigh these factors. The results also highlight the crucial role of parafoveal preview in efficient reading.

I. Introduction

Word skipping, the phenomenon in which a reader's eyes move past a word without directly fixating on it, has long been recognized as a fundamental feature of fluent reading. In skilled first-language (L1) reading, approximately one-third of words are skipped during reading in alphabetic languages (Rayner, 1998), and the likelihood of skipping has been argued to be systematically constrained by both visual and linguistic factors. Words that are short, high in frequency, or predictable from their preceding context tend to be skipped more often than long, infrequent, or unpredictable words (Rayner et al., 2011; Brysbaert et al., 2005). Because skipping reflects the earliest stage of lexical access and contextual integration, it provides a critical window into the mechanisms of visual word recognition and cognitive control during reading. Recent research has been extended to the second-language (L2) contexts, where skipping behavior serves as a sensitive indicator of processing efficiency and parafoveal processing capacity.

The phenomenon of word skipping is closely tied to parafoveal processing, the extraction of information from words just outside the reader's current fixation. The parafoveal region, typically spanning one to five degrees of visual angle to the right of the central fovea in left-to right languages, provides limited visual acuity but nonetheless enables readers to obtain orthographic, phonological, and even semantic cues from upcoming words (Liversedge & Findlay, 2000). This process, commonly referred to as parafoveal preview, plays a decisive role in whether an upcoming word will be directly fixated or skipped.

Several theoretical accounts have been advanced to explain the mechanisms through which parafoveal preview influences skipping. The Autonomous Oculomotor Scanning Strategy (McConkie & Yang, 2003) attributes skipping primarily to low-level oculomotor and visual mechanisms, suggesting that eye movements follow mechanical regularities largely independent of ongoing language comprehension. In contrast, the Language Control View (Brysbaert et al., 2005) posits that eye movement decisions, including skipping, are tightly linked to the progress of linguistic processing: words are skipped because they have been recognized or sufficiently predicted from context. This idea is reflected in major computational models such as E-Z Reader (Reichle et al., 1998) and SWIFT (Engbert et al., 2005), which posit that skipping occurs when a word in the parafovea reaches a sufficient degree of lexical activation before the eyes have moved to it. The resulting cancellation of a planned eye movement reflects efficient cognitive-linguistic processing. A third account, the Extended Optimal Viewing Position (EOVP) model (Brysbaert & Vitu, 1998), proposes that skipping is often an "educated guess", reflecting an experience-based estimation that the skipped word can either be easily recognized parafoveally or safely inferred from context.

Various studies have investigated predictors related to skipping to evaluate these competing theoretical accounts. A set of major factors that govern word skipping has been identified across languages and populations. In terms of oculomotor factors, word length consistently emerges as the strongest predictor, with shorter words being skipped more frequently than longer ones (Rayner, 1979; Schotter & Dillon, 2025). The launch site distance of the saccade, which is defined as the distance between the current fixation point and the beginning of the upcoming word, also modulates skipping probability, since shorter launch sites distance

improve the quality of parafoveal preview (McDonald & Shillcock, 2003; Brysbaert et al., 2005). On the linguistic side, word frequency and predictability play key roles. High-frequency words are recognized more quickly and thus are skipped more often (Staub, 2015; Esteve et al., 2024). Likewise, words that are more predictable in context show higher skipping rates than unpredictable words (Balota et al., 1985; Fernandez & Allen, 2025). Neurocognitive evidence of parafoveal N400 effects further supports the notion that readers can pre-activate semantic information before fixation, reflecting the interplay of expectation and visual processing during fluent reading (Milligan, 2024).

Although most of the research on word skipping has focused on L1 readers, recent work has extended this inquiry to second-language (L2) reading, offering new insights into bilingual language processing. Esteve et al. (2024) analyzed a large cross-linguistic dataset from the Multilingual Eye-Movement Corpus (MECO) and found that L2 word skipping is influenced not only by word length and frequency, but also by individual and cross-language differences. Readers with higher L2 proficiency and those from L1 backgrounds with higher average skipping rates exhibited more frequent skipping in English, suggesting a transfer of reading "habits" from L1 to L2. In addition, Fernandez and Allen (2025) proposed the *Reduced Capacity for Parafoveal Processing* (ReCaPP) hypothesis to explain quantitative differences in skipping between L1 and L2 English readers. Their study replicated the classic effects of word length and predictability using controlled materials (Slattery & Yates, 2018) and found that L2 English readers, while showing the same direction of effects as L1 readers, skipped less overall and had longer gaze durations. Importantly, the magnitude of these effects was modulated by proficiency and quality of lexical representation: higher-proficiency readers with stronger lexical

representations exhibited greater skipping rate and shorter gaze durations. The authors concluded that L1 and L2 readers share the similar mechanisms of predictability but differ in efficiency due to reduced parafoveal processing in L2 reading.

Despite these advances, significant gaps remain in our understanding of how parafoveal processing drives skipping in L2 reading. Previous studies have established the effects of word length, frequency, and predictability, but none of them have examined where and how these effects originate within the parafoveal preview window. This means that the interaction between parafoveal preview quality and other oculomotor and linguistic factors remains poorly understood. Moreover, the relative weighting of visual versus linguistic information may differ systematically between L1 and L2 readers, reflecting differences in lexical accessibility, reading efficiency, and attentional control.

Using a large corpus of naturalistic eye-movement data from both L1 and L2 English readers, the present study addresses these issues by directly quantifying parafoveal preview quality and modeling how oculomotor and linguistic factors jointly shape first-pass word skipping within the parafoveal preview window. By integrating measures such as launch site distance, word length, word frequency, and contextual predictability calculated from GPT based surprisal, this research aims to explain how visual and linguistic processes interact during reading and how these interactions differ across readers with different language backgrounds. Furthermore, the study addresses a notable research gap in the literature by focusing specifically on first-pass skipping, that is, skipping at the initial encounter with the word before moving past it in the text. This type of skipping behavior has been largely overlooked in previous corpus

research despite its importance for understanding real-time reading processes. Through this multifaceted approach, the study seeks to investigate and model the mechanisms underlying the differences between L1 and L2 word skipping and contribute to a more unified account of eye-movement control in reading.

Therefore, the research questions of the current study are twofold:

(1) How do oculomotor and linguistic factors influence first-pass skipping during L1 and L2 English reading?

(2) How does the first-pass skipping differ between L1 and L2 English readers?

II. Method

1. Defining the Major Predictors

1.1 Oculomotor Factors

Two oculomotor predictors, word length and launch site distance, were defined to capture spatial and visual constraints shaping parafoveal preview quality during reading. These variables reflect the low-level oculomotor mechanisms underlying saccade planning and skipping.

Word length was defined as the total number of letters in the word, including adjacent punctuation marks (termed *object length*). This definition ensures that all visual elements contributing to spatial extent are captured, since punctuation and word boundaries influence saccade targeting and fixation landing positions (Schotter & Dillon, 2025). Longer words extend further into the parafovea, decreasing the likelihood that their identity can be extracted without

direct fixation. Accordingly, shorter words are expected to be skipped more often, consistent with well-established perceptual span constraints (Rayner, 1979; Brysbaert et al., 2005).

Launch site distance was measured to quantify parafoveal preview quality for both fixated and skipped words. Specifically, launch site distance was defined as the number of letter spaces between the launch site on the current word (*word n*) and the initial letter of the subsequent word (*word n+1*), with each letter corresponding to a uniform spatial width. Because the text was displayed such that three letters equal approximately one degree of visual angle, launch site distance can be interpreted as a direct index of parafoveal preview quality.

Shorter launch site distances indicate that the upcoming word is closer to the reader's current fixation, providing a clearer parafoveal preview of the word and thus a higher probability of skipping. Conversely, longer launch sites degrade the parafoveal preview quality, reducing the chance of skipping (McDonald & Shillcock, 2003; Brysbaert et al., 2005; Milligan, 2024). By modeling this continuous metric, the present study captures the detailed parafoveal preview information available across successive fixations within the same line.

1.2 Linguistic Factors

Two linguistic variables, word frequency and predictability, were also included to represent lexical accessibility and contextual expectancy, respectively. Both variables capture higher-level cognitive-linguistic processes that can influence whether a word is fixated or skipped.

Word frequency was derived from the SUBTLEX$_{US}$ corpus (Brysbaert & New, 2009), which provides frequency based on naturalistic English texts. Word frequency values were log-transformed prior to analysis to normalize their skewed distribution. High-frequency words, being more easily recognized, are hypothesized to be skipped more often due to faster lexical access and stronger memory representations (Staub, 2015; Esteve et al., 2024).

Word predictability was estimated using GPT surprisal, a measure derived from the open-source GPT-2 model, which computes token-level probabilities based on a context window (Radford et al., 2019). Prior work has shown that the GPT-2 model correlates well with reading times in L1 English readers, indicating that this relatively smaller LLM (Large Language Model) is more suitable for capturing meaningful human-like contextual expectations compared to larger, state-of-the-art models (Berzak & Levy, 2023). A surprisal value was calculated for each word (w) as the negative logarithm of its conditional probability given the preceding context, as shown in (1).

(1) $S(w_i) = -\log P\left(w_i \mid w_1 \dots w_{\{i-1\}} \right)$

(Levy, 2008)

Lower surprisal values correspond to higher predictability (i.e., more expected words). More predictable words are argued to be skipped more often than less predictable words.

By combining frequency-based lexical accessibility and surprisal-based contextual expectancy, the present study incorporated both bottom-up and top-down components of linguistic processing. The interaction between these linguistic measures and oculomotor

predictors (word length, launch site distance) allows a more complete understanding of how cognitive and perceptual systems guide word skipping during reading.

2. Dataset

2.1 Description of the Current Eye-Tracking Data

The data were drawn from the Eye Movement, Reading, and Language Development (EMRLD) Corpus (Witzel & Witzel, 2025), an ongoing eye-tracking project at the University of Texas at Arlington (UTA) investigating reading processes in L1 and L2 English readers. The corpus includes detailed eye-movement records, text annotations, and individual difference measures. For the present study, data were analyzed from six expository passages covering a range of difficulty levels and topics in the natural and social sciences.

All texts were displayed in 18-point Courier New monospaced font, double-spaced, on a 19-inch CRT monitor located approximately 60 cm from the participant's eyes. At this distance, three letters spanned about one degree of visual angle, and the fixed-width font ensured consistent spacing for parafoveal preview. Each participant read six passages divided into 3 sections, each followed by a comprehension question to ensure engagement and understanding.

Eye movements were recorded using an EyeLink 1000 tracker (SR Research) at a 1000 Hz sampling rate. Only the right eye was tracked, though participants viewed the text binocularly. Calibration was performed before each passage using a nine-point grid, and drift correction was applied before each segment. Data preprocessing and fixation mapping were

carried out in SR Research DataViewer. Fixations shorter than 80 ms were excluded, and fixations associated with tracker loss or line misalignment were removed.

Each word in the dataset was coded for linguistic properties (word frequency, GPT surprisal, part of speech, and punctuation), visual-spatial features (word length, line number, word position, and launch site distance), and eye-movement outcomes (skipping, fixation durations, gaze durations). The dependent variable of interest in this study was first-pass skipping probability, defined as whether a word was fixated or skipped during the initial pass through the line. This high-resolution dataset enabled the modeling of how oculomotor and linguistic variables influence skipping across both L1 and L2 groups.

2.2 Participant Data

A total of 206 participants were drawn from the EMRLD corpus; 190 remained after exclusion for data quality. The final sample included 88 L1 English readers and 102 L2 English readers. All participants were university students, faculty, or visiting scholars at UTA. The L1 group consisted primarily of native speakers of American English who had acquired English before age five and used it as their dominant language. The group was demographically diverse, including early bilinguals with various secondary languages (e.g., Spanish, Hindi, Vietnamese).

The L2 group was linguistically and typologically diverse, representing 23 first languages from seven language families, including Indo-Aryan (Hindi, Bengali, Marathi, Nepali, Urdu), Dravidian (Tamil, Telugu, Kannada), Sino-Tibetan (Chinese), Iranian (Persian), Turkic (Azeri,

Turkish), Afro-Asiatic (Arabic, Amharic), and Romance (Spanish). Over 90% of L2 participants came from non-Latin script backgrounds.

III. Data Analysis

To address the research questions concerning how oculomotor and linguistic factors contribute to first-pass word skipping in L1 and L2 English reading, a logistic linear mixed-effects model (*glmer* in R) was developed to predict the binary dependent measure, i.e., first-pass skipping (1 = skipped; 0 = fixated). The present model therefore estimates the log-odds that a word will be skipped as a function of both oculomotor and linguistic properties of the word, as well as reader group (L1 or L2).

The model incorporated three categories of predictors: oculomotor factors, linguistic factors, and reader group. Oculomotor factors were included because they index the perceptual and positional constraints under which the eyes must operate during reading. These predictors comprised word length and launch site distance. Word length, measured in letter spaces (including punctuation), is known to strongly affect skipping probability. Launch site distance, the number of letter spaces between the reader's current fixation and the beginning of the next word, served as an index of parafoveal preview quality. Shorter distances enable greater parafoveal extraction of orthographic and lexical features due to higher visual acuity.
The model also included two linguistic predictors: word frequency, based on $SUBTLEX_{US}$ database, and word predictability, computed as GPT surprisal. Word frequency reflects long-term lexical familiarity and is a well-established determinant of ease of identification. Higher

predictability (i.e., low surprisal) reflects the extent to which the current word is anticipated given the preceding context; words that are highly expected require fewer visual resources for identification and thus are more likely to be skipped.

To evaluate group-level differences, reader group (L1 vs. L2 English) was included as a fixed effect. Sum coding was implemented such that the coefficients for group-related predictors reflect the probability for L1 readers. Critical interaction terms were included to examine potential group differences in sensitivity to oculomotor and linguistic information. Specifically, the model included interactions between group and each predictor, word length, launch site distance, word frequency, and word predictability, to assess whether L1 and L2 readers rely on these variables in distinct ways.

Additionally, the model included cross-factor interactions (e.g., word length × launch site distance; word frequency × launch site distance). Of particular interest were interactions involving launch site distance since such effects bear on the question of whether the availability of high-quality parafoveal preview modulates the influence of other variables. Such interactions are essential for evaluating the hypothesis that L1 and L2 readers may differ not only in their overall skipping rates, but also in the weighting of oculomotor versus linguistic information under varying parafoveal preview conditions.

Because the dataset consisted of multiple word-level observations per participant and per item, the model included random intercepts for participants and items, allowing the baseline

skipping rate to vary across individuals and lexical targets. Random slopes were included when justified to capture individual variability in sensitivity to the predictors and their interactions.

The final model converged without warnings and yielded stable, interpretable coefficients. Overall, the analytic framework allowed for a fine-grained assessment of the contributions of oculomotor and linguistic factors to skipping behavior and enabled direct comparisons between L1 and L2 readers within a unified modeling architecture.

IV. Results

Table 1 shows the output of the logistic linear mixed model, which revealed significant main effects and interactions that collectively illustrate how oculomotor and linguistic information guide word skipping during first-pass reading. The findings also underscore systematic differences between L1 and L2 readers in their reliance on these factors.

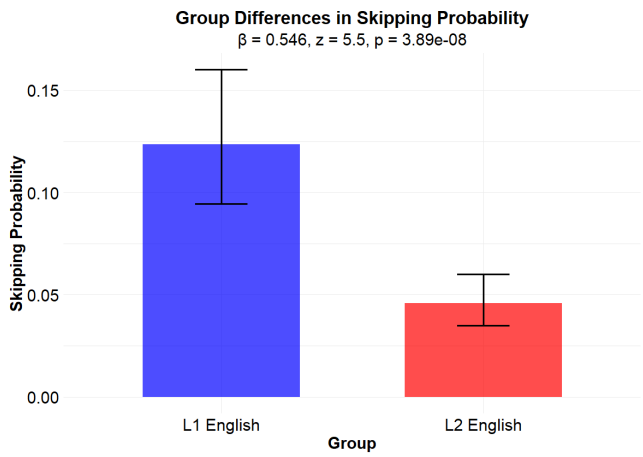
Table 1. Fixed Effects Structure of the Logistic Linear Mixed Model

Predictor	*Estimate*	*SE*	*z value*	*p value*
(Intercept)	-2.59367	0.11375	-22.801	< 2e−16 ***
Group (L1 vs. L2)	0.54551	0.09926	5.496	3.89e−08 ***
Word Length	-1.75656	0.07050	-24.914	< 2e−16 ***
Launch Site Distance	-1.22493	0.05013	-24.433	< 2e−16 ***
Word Frequency	0.33196	0.05958	5.572	2.52e−08 ***
Word Predictability	-0.22284	0.03815	-5.841	5.18e−09 ***
Group × Word Length	0.20622	0.04283	4.814	1.48e−06***
Group × Launch Site Distance	-0.13045	0.04468	-2.920	0.003505 **
Group × Word Frequency	0.14421	0.02661	5.420	5.97e-08 ***
Word Length × Launch Site	0.11520	0.03055	3.771	0.000162 ***
Word Frequency × Launch Site	-0.18607	0.03594	-5.310	1.10e−07 ***
Word Predictability × Launch Site	0.05434	0.01948	2.789	0.005284 **
Word Length × Word Frequency	-0.29432	0.04516	-6.517	7.17e-11 ***
Group × Word Length × Launch Site	-0.04963	0.01354	-3.666	0.000246 ***
Group × Word Frequency × Launch Site	0.07900	0.02265	3.488	0.000487 ***
Group × Word Length × Word Frequency	0.11290	0.02113	5.343	9.12e-08 ***
Word Length × Word Frequency × Launch Site	-0.08059	0.02104	-3.829	0.000129 ***

1. Overall Group Differences

There was a significant main effect of reader group, as shown in Fig. 1. L1 readers demonstrated a higher baseline probability of skipping words than L2 readers ($\beta = 0.546$, $p < .001$). This difference persisted even after controlling oculomotor and linguistic predictors, suggesting a fundamental difference in reading strategy: L1 readers engage in more efficient parafoveal processing and lexical access, allowing them to skip words more readily. In contrast, L2 readers tend to adopt a more cautious strategy, often fixating words that L1 readers would skip, likely due to lower lexical familiarity, reduced visual efficiency, or both.

Fig.1 Group Differences



2. Oculomotor Predictors

2.1 Word Length

Consistent with previous findings in eye-movement research, word length showed a strong negative effect on skipping probability (β = –1.757, $p < .001$), as shown in Fig. 2a. Shorter words were significantly more likely to be skipped. Notably, the group × word length interaction (β = 0.206, $p < .001$) in Fig. 2b showed that the effect of length was more pronounced for L2 readers. As word length increased, L2 readers showed a sharper decline in skipping probability relative to L1 readers. This pattern suggests that L2 readers may be more dependent on direct fixation for accurate lexical access.

Fig. 2a Word Length Effect

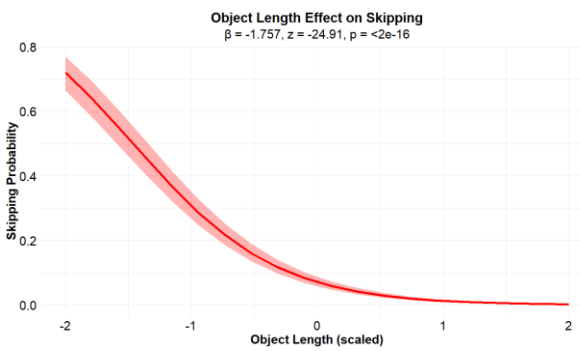
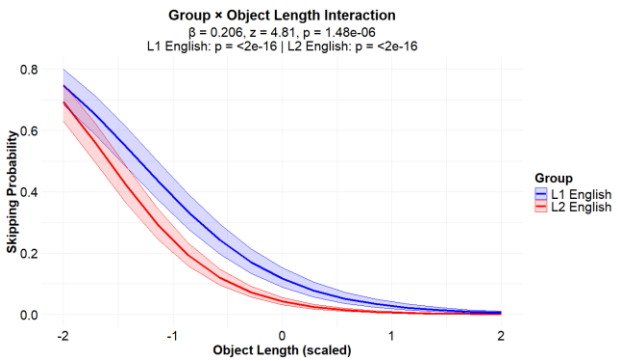


Fig. 2b Group × Word Length



2.2 Launch Site Distance

Launch site distance also produced a robust negative effect (β = –1.225, $p < .001$), as shown in Fig. 3a, with shorter distances associated with higher skipping rates. This effect reflects the importance of parafoveal preview: when the fixation is closer to a subsequent word, the visual system can extract more information from that word, increasing the likelihood it will be skipped.

The significant group × launch site distance interaction (β = –0.130, $p < .01$) in Fig. 3b indicated that L2 readers benefited less from short launch site distances than L1 readers. This

finding is consistent with the *Reduced Capacity for Parafoveal Processing* (ReCaPP) hypothesis, which argues that L2 readers extract less useful information from the parafoveal region and therefore cannot fully capitalize on preview opportunities. For L2 readers, the advantage of short launch site distances is diminished relative to L1 readers, further contributing to their lower overall skipping rates.

Fig. 3a Launch Site Distance Effect

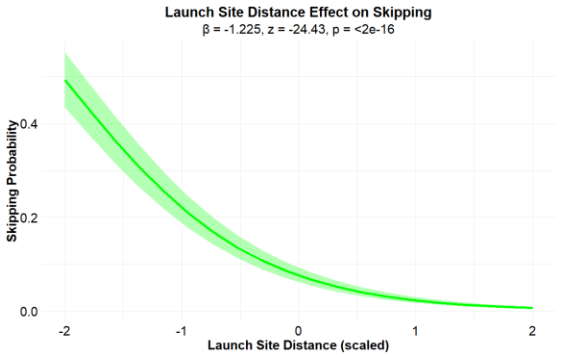
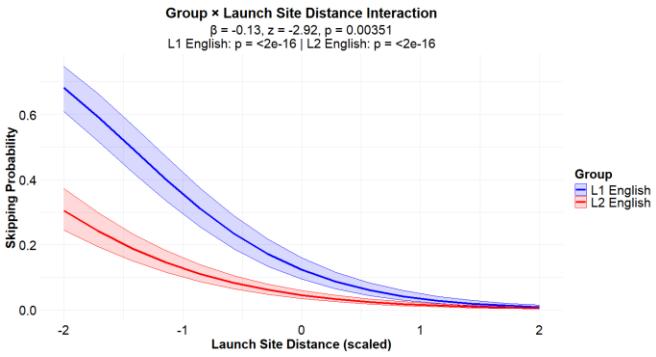


Fig. 3b Group × Launch Site Distance



3. Linguistic Predictors

3.1 Word Frequency

Word frequency showed a significant positive effect ($\beta = 0.332$, $p < .001$), as shown in Fig. 4a, such that high-frequency words were more likely to be skipped. The significant group × frequency interaction ($\beta = 0.144$, $p < .001$) in Fig. 4b demonstrated that frequency effects were stronger for L1 readers. This effect suggests that L1 readers are more adept at utilizing lexical familiarity to guide skipping, whereas L2 readers show a reduced frequency effect, likely reflecting less robust lexical representations or slower lexical access.

Fig. 4a Word Frequency Effect

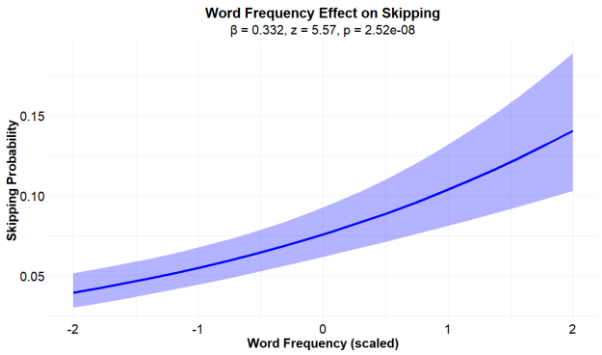
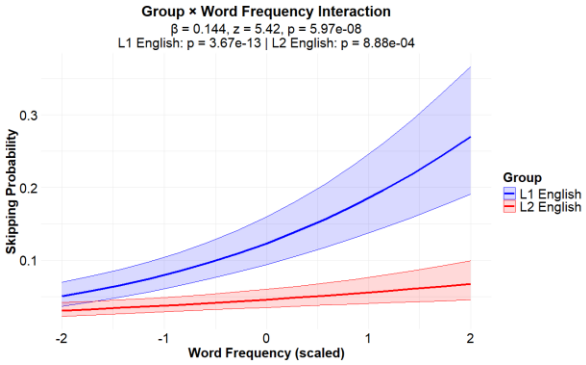


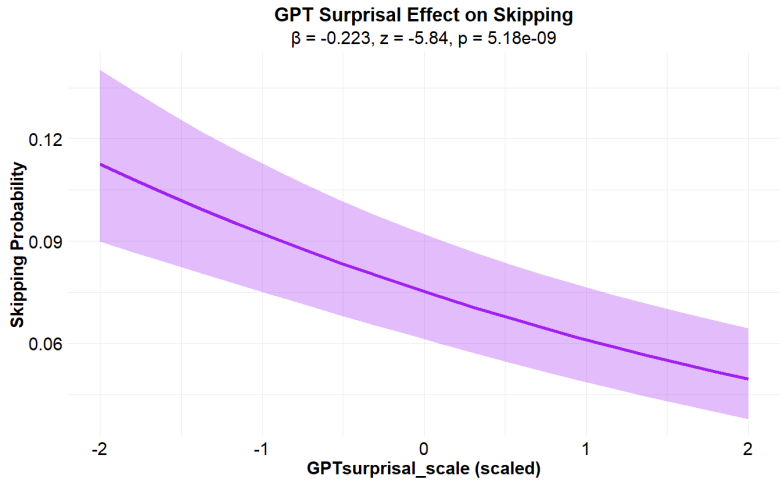
Fig. 4b Group × Word Frequency



3.2 Word Predictability (GPT Surprisal)

Surprisal exhibited a negative main effect (β = –0.223, $p < .001$), as shown in Fig. 5, indicating that words that were more predictable given prior context were more likely to be skipped. This finding aligns with expectation-based models of reading (Levy, 2008), in which readers pre-activate likely upcoming words and therefore require less visual information to identify them. Importantly, no significant group × word predictability interaction emerged. Both L1 and L2 readers appeared to rely on contextual predictability to a comparable extent. This result is notable because it suggests that, despite differences in lexical proficiency or parafoveal efficiency, L2 readers maintain relatively intact predictive processing mechanisms, at least at the level of contextual anticipation captured by language models such as GPT2.

Fig. 5 Word Predictability Effect



4. Interactions Involving Parafoveal Preview

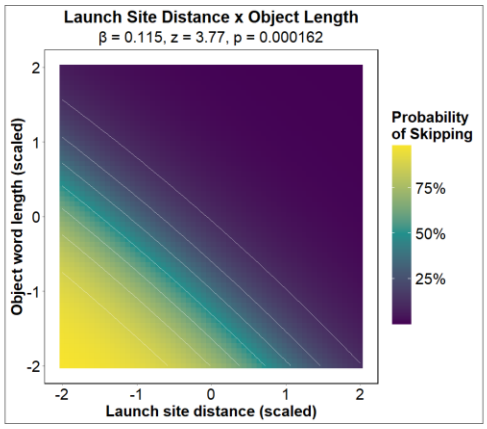
The statistical model further demonstrated that the influence of oculomotor and linguistic factors is modulated by parafoveal preview quality as indexed by launch site distance.

4.1 Interactions Between Word Length and Parafoveal Preview

There was a significant interaction between word length and launch site distance ($\beta = 0.115$, $p < .001$), as shown in Fig. 6a, indicating that the effect of word length on skipping probability is modulated by parafoveal preview quality. The positive coefficient shows that the inhibitory effect of word length on skipping becomes weaker as launch site distance increases. When the launch site is close (i.e., parafoveal preview quality is high), word length exerts a strong negative effect: short words are skipped more often, while long words are skipped relatively less often. As launch site distance increases and parafoveal preview degrades, the difference between short and

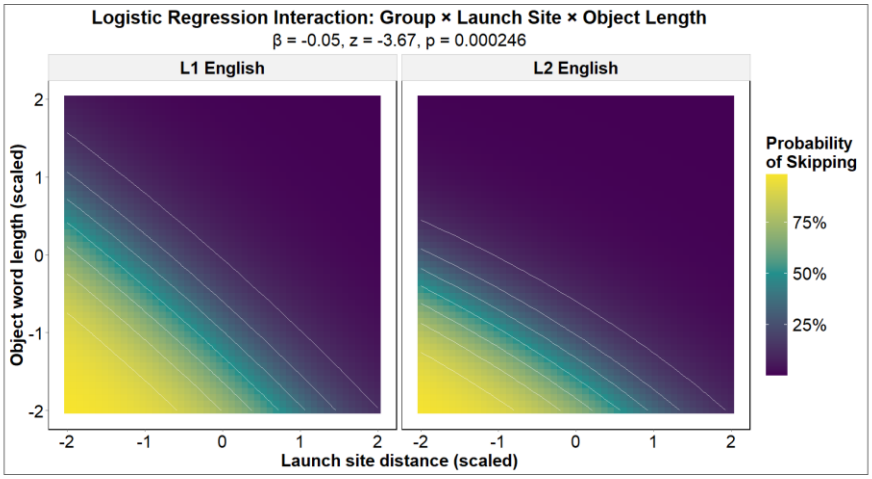
long words narrows. This pattern suggests that high-quality parafoveal preview amplifies the length-based asymmetry in skipping, whereas poor preview suppresses it.

Fig. 6a Launch Site Distance × Word Length



Crucially, this interaction was further qualified by a three-way interaction involving group (L1 vs. L2) ($\beta = -0.050$, $p < .001$), as shown in Fig. 6b. The negative coefficient indicates that L2 readers show a larger word length × launch site distance interaction than L1 readers. In other words, skipping behavior in L2 readers is more strongly shaped by the combination of word length and parafoveal preview quality. When parafoveal preview is strong, L2 readers demonstrate an especially sharp increase in skipping probability for short words relative to long words; when preview is weak, skipping drops more steeply for L2 readers across all word lengths. This heightened dependence on parafoveal preview suggests that L2 readers require stronger perceptual support to make efficient skipping decisions and that the absence of high quality parafoveal information disproportionately reduces their willingness to skip words. The pattern indicates that oculomotor and visual constraints play an amplified role in shaping the early reading behavior for L2 readers.

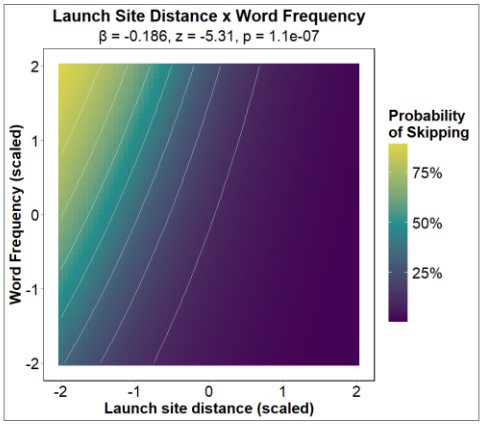
Fig. 6b Group × Launch Site Distance × Word Length



4.2 Interactions Between Word Frequency and Parafoveal Preview

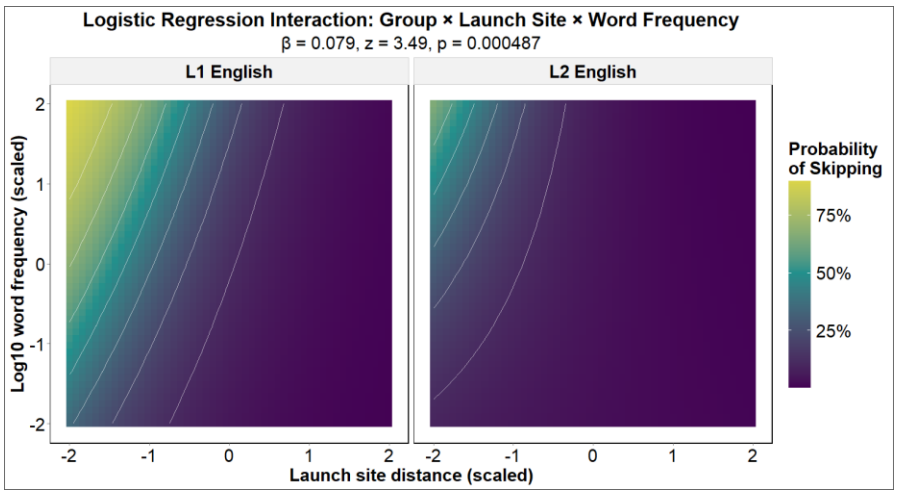
A significant interaction also emerged between word frequency and launch site distance ($\beta = -0.186$, $p < .001$), as shown in Fig. 7a, indicating that the influence of lexical familiarity on skipping probability depends significantly on the parafoveal preview quality. The negative coefficient shows that the effect of word frequency becomes weaker as launch site distance increases. When readers begin their saccade from a short launch site distance, high-frequency words receive a strong skipping boost because readers can extract enough parafoveal information to activate their familiar lexical representations. As launch site distance increases and preview quality diminishes, this advantage declines: readers become less able to use word frequency information, causing skipping rates for high- and low-frequency words to converge. Thus, frequency-driven skipping is enhanced under optimal perceptual conditions and reduced when parafoveal input is degraded.

Fig. 7a Launch Site Distance × Word Frequency



This interaction was further moderated by reader group, as reflected in the significant group × word frequency × launch site distance effect ($\beta = 0.079$, $p < .001$), as shown in Fig. 7b. Importantly, the positive coefficient should be interpreted relative to the negative coefficient of the two-way interaction in Fig. 7a. A positive value here indicates that the negative frequency × launch site distance interaction is more negative for L2 readers and less negative for L1 readers. In other words, the word frequency and parafoveal preview interaction is stronger for L2 readers. This means that L2 readers show a steeper reduction in the frequency effect as parafoveal preview worsens, indicating that lexical familiarity becomes less useful for them more quickly than for L1 readers. Conversely, it is only at very short launch site distances that L2 readers exhibit skipping advantage for high-frequency words. Their reliance on lexical familiarity is therefore particularly dependent on the availability of optimal parafoveal preview.

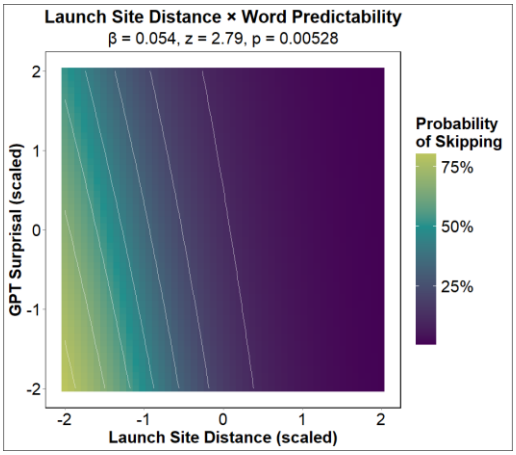
Fig. 7b Group × Launch Site Distance × Word Frequency



4.3 Interactions Between Word Predictability and Parafoveal Preview

The model also revealed a significant interaction between word predictability and launch site distance, indicating that contextual expectancy effects on skipping depend systematically on parafoveal preview quality. The positive coefficient for the interaction ($\beta = 0.054$, $p < .01$), as shown in Fig. 8, demonstrates that the influence of predictability becomes weaker as launch site distance increases and stronger when launch site distance is short.

Fig. 8 Launch Site Distance × Word Predictability

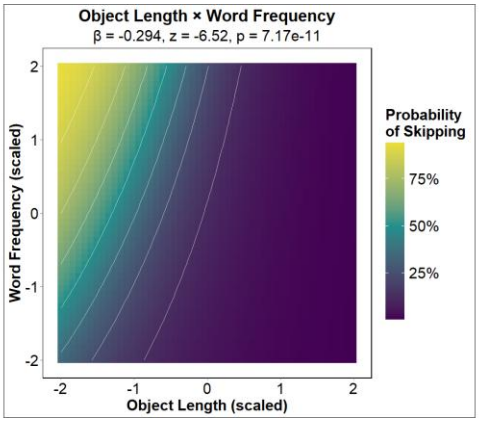


This pattern suggests that readers rely more heavily on contextual expectations when the upcoming word is close enough to the point of fixation to afford a relatively clear parafoveal preview. Under these conditions, the parafoveal system can combine partial visual information with strong contextual cues to support confident skipping. In contrast, when the launch site is farther from the word and visual input is degraded, the system is less able to make use of contextual predictability, thereby reducing the predictability-based advantage in skipping. Overall, the word predictability × launch site distance interaction indicates that contextual information in word skipping is constrained by perceptual conditions: predictability shows its strongest influence when parafoveal preview is of sufficiently high quality to support the integration of visual and contextual information. Also importantly, there is no significant group difference between L1 and L2 readers.

5. Interactions involving Word Length and Word Frequency

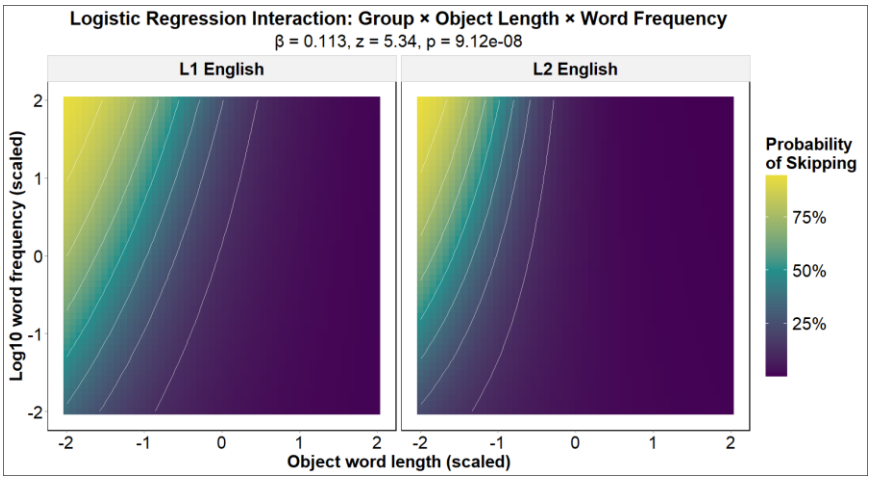
The significant word length × word frequency interaction ($\beta = -0.294$, $p < .001$) in Fig. 9a showed that the effect of word frequency was attenuated as word length increased. High-frequency words were skipped more often when they were short, consistent with greater parafoveal accessibility. This frequency advantage diminished for longer words, suggesting that lexical familiarity is less useful when visual demands increase and parafoveal extraction becomes more difficult.

Fig. 9a Word Length × Word Frequency



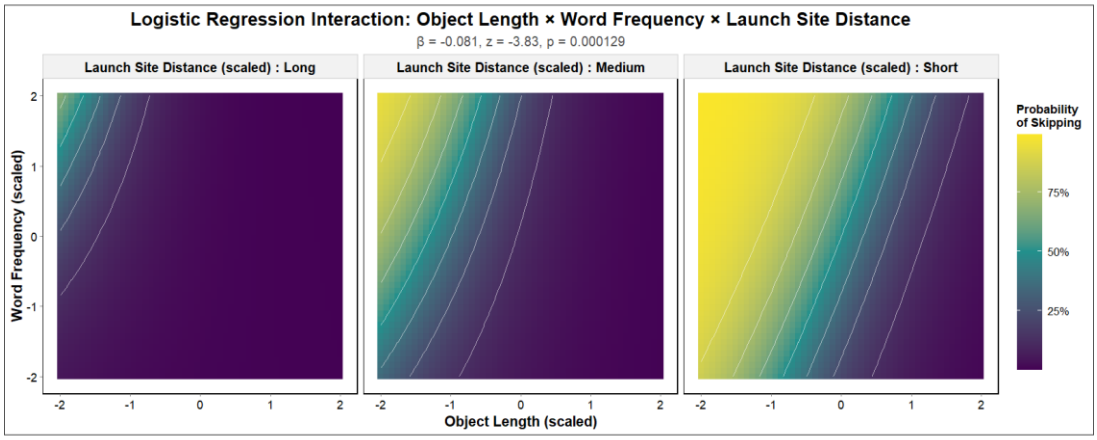
This pattern was further moderated by reader group. The group × word length × word frequency interaction (β = 0.113, $p < .001$), as shown in Fig. 9b, indicated that the length-related reduction of the frequency effect was more pronounced for L2 readers. L2 readers showed a strong frequency advantage for short words but a steeper decline in this effect for longer words. By contrast, L1 readers maintained a more stable frequency effect across word lengths. These results suggest that L2 readers' ability to use lexical familiarity during parafoveal processing is more sensitive to increases in visual complexity.

Fig. 9b Group × Word Length × Word Frequency



A significant word length × word frequency × launch site distance interaction (β = –0.081, $p < .001$), as shown in Fig. 10, further demonstrated that the strength of the frequency effect depended on parafoveal preview quality. The reduction of the frequency effect for longer words was amplified at greater launch site distances, where parafoveal information is weaker. Conversely, with shorter launch site distances, the frequency advantage for short words was particularly strong. This pattern underscores that both word length and parafoveal preview jointly constrain the extent to which readers can rely on lexical familiarity when making skipping decisions.

Fig. 10 Word Length × Word Frequency × Launch Site Distance



Across these complex interactions, a consistent pattern emerges: the influence of word frequency to skipping is strongly conditioned by oculomotor factors. Frequency effects are most robust when perceptual conditions are favorable, specifically when words are short and parafoveal preview quality is high, and become weaker when visual complexity increases or parafoveal information is limited. L2 readers are especially sensitive to these constraints: they exhibit a

strong frequency advantage for short words under good parafoveal preview conditions, but this advantage declines sharply for longer words and at greater launch site distances. In contrast, L1 readers show a more stable frequency effect across variations in word length and parafoveal preview.

6. Summary

Taken together, the results support a hybrid account of word skipping in which oculomotor and linguistic factors jointly determine skipping probability. L1 readers exhibit higher skipping rates and make more efficient use of both parafoveal preview and lexical familiarity. L2 readers, in contrast, rely more heavily on low-level cues such as word length and show reduced sensitivity to frequency, reflecting differences in lexical access and parafoveal processing capacities. Nonetheless, predictability effects are preserved across groups, suggesting that expectation-based processes remain robust in overall reading. These findings contribute to a more comprehensive understanding of the mechanisms underlying reading proficiency and cross-linguistic differences in eye-movement behavior.

V. Discussion

The present study examined how oculomotor and linguistic factors jointly shape first-pass word skipping in L1 and L2 English readers. Using a logistic linear mixed-effects model, the analysis revealed robust main effects and interactions across predictors, as well as systematic differences between L1 and L2 readers. The discussion below synthesizes these findings, beginning with the

major predictors and their interactions, then moving toward broader theoretical and pedagogical implications.

1. Effects of Major Predictors and Group Differences

A central outcome of the study is the set of robust main effects for word length, lexical frequency, predictability, and launch site distance. Each predictor shaped skipping in expected directions, aligning with prior work demonstrating that shorter, more frequent, and more predictable words are easier to process and therefore less likely to require fixation (Rayner, 1998; Staub, 2015; Brysbaert & Vitu, 1998). Visual constraints also played a defining role: shorter launch site distances reliably increased skipping, reflecting the improved parafoveal preview available when saccades originate closer to the target word.

Crucially, these predictors did not operate independently. Their interactions, particularly those involving launch site distance, underscore that skipping emerges from dynamic integration rather than single-variable influence. For example, the greater skipping advantage for short words at short launch site distance suggests that visual accessibility enhances the effective use of lexical information. Similarly, frequency effects strengthened under favorable parafoveal preview conditions, reflecting the joint contribution of lexical automaticity and perceptual clarity.

The comparison between L1 and L2 readers further highlights how processing efficiency shapes reliance on different factors. L2 readers exhibited stronger effects of word length but

weaker effects of word frequency, indicating heavier dependence on oculomotor factors and reduced use of lexical familiarity. These group differences provide an important foundation for interpreting how parafoveal preview and lexical processing interact.

2. Parafoveal Preview Modulation and the Role of Launch Site Distance

Based on these group-specific patterns, the results underscore the central role of parafoveal preview in guiding skipping. Launch site distance was not merely a spatial parameter but a powerful modulator that amplified or constrained other factors. When readers approached a word from a short launch site distance with high-quality parafoveal preview, smaller word lengths, higher frequencies, and higher predictability converged to increase the likelihood of skipping.

The modulation of different effects by parafoveal preview also provides clear evidence for group differences. L2 readers exhibited a significantly smaller launch site effect (Group × Launch Site: $\beta = -0.130$, $p < .01$), suggesting reduced efficiency in integrating parafoveal information into saccade planning. However, their reliance on word length and lexical familiarity is highly dependent on the availability of optimal parafoveal preview indexed by short launch site distance. Even though they show reduced effect of parafoveal preview, L2 readers depend more strongly on optimal parafoveal preview conditions than L1 readers when lexical variables such as word length and frequency are favorable. Therefore, instead of *Reduced Capacities for Parafoveal Processing* (Fernandez & Allen, 2025), L2 readers exhibit more conditional and less automatized skipping behavior: they are more likely to skip short and high-frequency words when the parafoveal preview quality is good.

3. Differences Between L1 and L2 English Readers

These parafoveal preview-driven differences lead naturally to a broader comparison of L1 and L2 skipping strategies. L1 readers demonstrated more balanced integration of visual, lexical, and contextual information, consistent with a high degree of fluency that allows them to coordinate multiple sources of information rapidly (Rayner, 1998). Their stable effects of frequency and predictability indicate that linguistic factors play a substantial role in driving saccade decisions.

In contrast, L2 readers showed a more bottom-up reading pattern, marked by conservative and less automatized processing typical of L2 reading. They showed heightened sensitivity to word length and reduced influence of word frequency. Overall, these results illustrate that while L2 readers rely on many of the same cues as L1 readers, they weigh the factors differently due to more conditional processing strategies. These complementary findings set the stage for articulating a theoretical account that incorporates both linguistic and oculomotor components.

4. A Hybrid Theoretical Account of Skipping Behavior

In sum, the results support a hybrid theoretical account in which skipping emerges from the interaction of oculomotor constraints and linguistic processing. Neither component alone can fully explain the observed patterns. The strong effects of word length and launch site distance highlight the continued influence of perceptual constraints, while the significant contributions of frequency and predictability demonstrate the role of lexical familiarity and expectation in

shaping saccade planning. Importantly, the interactions among predictors indicate that visual and linguistic processes do not operate in isolation; instead, they work jointly to optimize reading efficiency.

The contrasting strategies of L1 and L2 readers further reinforce this hybrid view. Both groups rely on the same underlying mechanisms, but their relative weighting differs according to proficiency. L1 readers, equipped with automatized lexical knowledge and efficient parafoveal preview extraction, rely heavily on linguistic cues. L2 readers, in contrast, compensate for weaker lexical automaticity by depending on general visual cues such as word length and good parafoveal preview. These findings illustrate how the cognitive architecture of reading adapts to individual proficiency levels and how the balance of oculomotor and linguistic factors shifts as readers gain experience.

5. Pedagogical Implications

These theoretical insights carry meaningful implications for L2 reading instruction. The reduced influence of lexical frequency and the importance of parafoveal preview on L2 skipping suggest that instruction should prioritize developing automaticity in word recognition, rather than focusing solely on vocabulary size or definitional knowledge.

Because L2 readers demonstrated weaker integration of linguistic and parafoveal information, pedagogical approaches that strengthen predictive processing, such as cloze task (filling the blanks), may help learners build stronger anticipatory mechanisms. Additionally,

training that targets perceptual span and saccade efficiency, such as speeded reading, may enhance parafoveal processing and help L2 readers integrate linguistic cues with visual information more effectively.

In conclusion, the study provides clear evidence that word skipping is governed by a combination of oculomotor and linguistic processes. The differential weighting of these processes between L1 and L2 readers highlights the importance of considering both processing efficiency and linguistic expertise in models of reading. Future research should continue to examine how these systems interact, particularly across proficiency levels and diverse linguistic backgrounds.

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